Long-Term Comparison Between Perindopril and Nifedipine in Normotensive Patients With Type 1 Diabetes and Microalbuminuria

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• The aim of this study is to compare the efficacy of an angiotensin-converting enzyme inhibitor with a dihydropyridine calcium channel blocker in preventing progression to macroalbuminuria and/or a decline in renal function in normotensive patients with type 1 diabetes and microalbuminuria. Forty-two patients were randomized to treatment with either perindopril, slow-release nifedipine, or placebo. In the first 3 months, drug dosage was titrated to achieve a decrease in diastolic blood pressure of at least 5 mm Hg. Thirty-three patients had a minimum of 24 months' data, and 25 patients were followed up beyond 36 months (mean, 67 ± 4 months). Patients were studied every 3 months and at the end of the treatment period; those who remained normotensive discontinued therapy and were followed up for an additional 3 months. Baseline geometric mean albumin excretion rates (AERs) were as follows: perindopril, 66 μg/min; nifedipine, 59 μg/min; and placebo, 66 μg/min. During the first 3 years, 7 of the perindopril-treated but none of the placebo or nifedipine-treated patients reverted to normoalbuminuria ($P < 0.01$). Median AERs at 3 years of treatment in each group were 23 μg/min for perindopril, 122 μg/min for nifedipine, and 112 μg/min for placebo patients ($P < 0.01$). In patients with more than 3 years' follow-up, median AERs decreased by 45% in the first year and then stabilized in the perindopril group, but increased by 17.6% in the nifedipine group and 27.6% in the placebo group ($P < 0.03$) in the first year, then increased progressively. In these same patients, there was a significant decline in glomerular filtration rate in the nifedipine group (−7.8 ± 1.8 mL/min/1.73 m²/y), but not in the other two groups (perindopril, −1.0 ± 1.2 mL/min/1.73 m²/y; placebo, −1.3 ± 1.1 mL/min/1.73 m²/y; $P = 0.004$). At the end of the study, cessation of treatment for 3 months was associated with a doubling of AERs in the perindopril-treated group, but no change in the other two groups ($P < 0.001$). In conclusion, long-term perindopril therapy is more effective than nifedipine or placebo in delaying the progression of diabetic nephropathy and reducing AER to the normoalbuminuric range (<20 μg/min) in normotensive patients with type 1 diabetes and microalbuminuria.

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INDEX WORDS: Type 1 diabetes; diabetic nephropathy; angiotensin-converting enzyme (ACE) inhibition; calcium channel blockade; microalbuminuria; glomerular filtration rate (GFR); albumin excretion rate (AER).

SEVERAL GROUPS have shown that angiotensin-converting enzyme (ACE) inhibitors are able to retard the development of overt diabetic nephropathy.[1,2] These studies have been performed in hypertensive patients with type 2 diabetes,[3] as well as in normotensive patients with type 1 diabetes.[2,4] It has been suggested that these renoprotective effects of ACE inhibitors may be partly independent of their effects on systemic blood pressure (BP).[5,6] However, the role of other antihypertensive agents, in particular, dihydropyridine calcium channel blockers (CCBs), in conferring similar effects on albuminuria has not been clarified, particularly in the normotensive context. The Melbourne Diabetic Nephropathy Study Group (MDNSG) previously reported similar efficacy of the ACE inhibitor perindopril and the dihydropyridine CCB nifedipine in preventing the development of macroalbuminuria in microalbuminuric patients with types 1 and 2 diabetes over a 12-month period.[7] However, that study did not include a placebo control group, and a significant proportion of patients had hypertension. Therefore, the MDNSG decided to perform a new study that included only normotensive microalbuminuric subjects with type 1 diabetes and three treatment arms: perindopril, sustained-release nifedipine, and placebo. While this study was in progress, several organizations published guidelines recommending that

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ACE inhibitors be considered routine first-line agents in normotensive microalbuminuric subjects with type 1 diabetes.[8,9] However, these guidelines were based on studies comparing ACE inhibitors with placebo and did not include a third group treated with another antihypertensive agent.[6] For this reason, it was decided to continue the study until a significant difference in albuminuria or glomerular filtration rate (GFR) emerged. We report here the findings for 33 patients followed up for 24 months.

SUBJECTS AND METHODS

Patient Eligibility

Five hospitals participated in the study, including three teaching hospitals of the University of Melbourne (Austin & Repatriation Medical Centre, St Vincent's, and Western Hospitals), Monash Medical Centre, and Adelaide Children's Hospital. Ethics committee approval was obtained from each center, and informed consent was obtained from all participants. This was a prospective, randomized, open, blinded end point evaluation study. Participants were unaware of the identity of the study drug. Drug dispensing, laboratory measurements, and statistical comparisons were performed independently of the treating doctor by staff who were masked to the randomization status of individual patients.

The criteria for inclusion in the study were patients aged 16 to 65 years classified as having type 1 diabetes for at least 5 years, with persistent ketonuria and weight loss on presentation, as well as insulin dependency within 1 month of diagnosis as defined by the National Diabetes Data Group[10]; microalbuminuria, defined as at least two of three measurements of albumin excretion rate (AER) between 20 and 200 μg/min (performed on overnight urine collections)[8]; supine BP of 160/90 mm Hg or less if aged 40 years or older or supine BP of 140/90 mm Hg or less if aged younger than 40 years. Exclusion criteria included: nondiabetic renal disease, significant renal impairment (serum creatinine level $\geq$ 0.2 mmol/L), hematuria, cardiac failure, hypertension, concurrent systemic disease, poor diabetic control (hemoglobin A_{1c} [HbA_{1c}] > 10%), serum potassium > 5 mmol/L, recurrent urinary tract infections, patients at high risk for pregnancy, and patients with another condition or therapy that might pose a risk to the patient or confound the results of the study.

Forty-two patients were recruited into the study. Thirty-three patients successfully completed at least 24 months of treatment (including 28 patients who completed 36 months). Nine patients failed to complete at least 24 months of treatment, including 4 patients who were lost to follow-up (2 patients, perindopril; 1 patient, nifedipine; 1 patient, placebo), 1 patient who died after 5 months on nifedipine therapy, 1 patient who developed lichen planus after 6 months of perindopril treatment, 1 patient who developed urticaria after 15 months of perindopril treatment, and 2 patients who withdrew because of time constraints (1 patient, nifedipine; 1 patient, placebo).

Randomization and Titration

Patients who met these criteria were randomized to treatment with either perindopril (Servier IRIS, Neuilly, France), nifedipine sustained release (Bayer, Wuppertal, Germany), or placebo (in the image of a perindopril tablet; Servier IRIS). Randomization was performed by a pharmacist who had no information about individual patients. Placebo dose was one tablet daily. The initial dose of perindopril was 2 mg administered once daily in the morning, then titrated at 2-week intervals to 4 mg, then 8 mg administered once daily in the morning, aiming for a reduction in supine diastolic BP of 5 mm Hg or greater. The initial dose of nifedipine sustained release was 10 mg twice daily, increasing to 20 mg twice daily, then 40 mg twice daily if well tolerated, aiming for a reduction in supine diastolic BP of 5 mm Hg or greater. Patients were seen at 2-week or monthly intervals until titration of their medication was completed, in all cases by 12 weeks after randomization.

End Points

The major end point of the study was urinary albumin excretion. Secondary end points included the development of macroproteinuria or reversal to the normoalbuminuric range. If hypertension, defined as three consecutive monthly readings of supine diastolic BP greater than 90 mm Hg and/or supine systolic BP greater than 140 mm Hg if aged 40 years or younger and/or supine systolic BP greater than 160 mm Hg if aged older than 40 years, developed or significant renal impairment, defined as a GFR less than 60 mL/min/1.73 m^2 and/or serum creatinine level greater than 0.20 mmol/L, was detected, the study agent was discontinued and the subject was allocated to either perindopril or nifedipine, with dose titration as outlined previously.

Study Protocol

After randomization and titration of dose, visits occurred every 3 months for a minimum of 2 years, and 28 of 33 patients had at least 3 years' follow-up. Twenty-five patients (9 patients, perindopril; 9 patients, nifedipine; 7 patients, placebo) were followed up beyond 3 years, with a mean treatment duration of 67.2 ± 4.1 months (range, 36 to 96 months). Patients then ceased therapy, and if they were still normotensive at the end of the study, they were followed up for a subsequent 3 months off treatment. Every 3 months, the following parameters were measured: office BP, body weight, two consecutive measurements of overnight AER, HbA_{1c} levels, plasma renin concentration, and urea and electrolyte values. Every 12 months, isotopic GFR was measured. At the end of the study, a final GFR was measured in all patients in whom the interval between the termination date and the previous GFR measurement was greater than 6 months. Ambulatory BP was determined at least once in each patient. Serum cholesterol and triglyceride levels were measured every 12 months.

Clinical Methods

Office BP was measured by means of an automated sphygmomanometer (Dinamap; Critikon, Tampa, FL) after 2 and 5 minutes in the supine and standing position at each

Table 1. Baseline Characteristics of Patients Who Did and Did Not Complete the Study

Variable	Completed (n = 33)	Did Not Complete (n = 9)
Age (y)	31 ± 2	35 ± 5
Diabetes duration (y)	16 ± 2	13 ± 2
Sex distribution (M/F)	17/16	5/4
Retinopathy (%)	56	60
Smoking (%)	34	40
Weight (kg)	71 ± 2	69 ± 3
HbA_{1c} (%)	8.9 ± 3	8.5 ± 0.8
Total cholesterol (mmol/L)	4.9 ± 0.2	4.9 ± 0.4
Triglycerides (mmol/L)	1.3 ± 0.1	1.0 ± 0.2
Systolic BP (mm Hg)	132 ± 2	136 ± 5
Diastolic BP (mm Hg)	78 ± 2	83 ± 2*
Albuminuria (μg/min)	64 ×/÷ 1.1	61 ×/÷ 1.2
GFR (mL/min/1.73 m^2)	125 ± 5	116 ± 18

NOTE. Data expressed as mean ± SEM, except for albuminuria (geometric mean ×/÷ tolerance factor).
* $P < 0.05$ compared with completed group.

visit. Adjustments in therapy were based on the 5-minute supine BP. Twenty-four–hour ambulatory BP monitoring was determined using a SpaceLabs 90207 device (Redmond, WA) in subjects during the active phase of treatment. Fasting plasma cholesterol and triglyceride levels were measured by enzymatic colorimetric methods, and plasma creatinine was measured by the modified Jaffe method using a Hitachi 747 autoanalyzer (Roche Diagnostics, Mannheim, Germany). Urinary albumin excretion was measured by radioimmunoassay.[11] Human albumin (Sigma, St Louis, MO) was iodinated by the chloramine-T method.[12] The detection limit of the assay was 15.6 ng/mL, and the interassay coefficient of variation was 7.3% (n = 37) at a concentration of 30 mg/L. All estimations of albumin concentration were performed at a single site on two urine collections per visit (Endocrinology Laboratory, Austin & Repatriation Medical Centre, Heidelberg, Australia). HbA_{1C} was measured by an automated high-performance liquid chromatography technique (Biorad Diamat, Richmond, CA). GFR was measured by a single-injection technique using the isotope technetium 99m diethylene triamine penta acetic acid.[13,14] Plasma renin concentration was measured by radioimmunoassay.[15]

Statistical Analysis

Differences in continuous variables (eg, age, AER) among the three treatment groups were compared by using one-way analysis of variance. Individual treatment groups were compared using Fisher's protected least significant difference (PLSD) post hoc test. Differences in categorical variables (eg, sex, regression to normoalbuminuria) were compared using chi-square analysis or Fisher's exact test when cell frequencies were small. Differences between patients completing and failing to complete the study were compared using Student's *t*-test for continuous variables and chi-square analysis or Fisher's exact test for categorical variables. Within-subgroup changes from baseline values were compared by using Student's *t*-test. Associations between BP and AER were tested by using Pearson's correlation. Data for AER were analyzed after logarithmic transformation. The change in AER was estimated using regression analysis of all available data up to 36 months and expressed as percentage of change in AER per year.[16] Data from the three patients in the placebo group who developed hypertension and were then randomized to active treatment were included in the placebo group as part of an intention-to-treat analysis.

In the 25 patients with data beyond 36 months, comparisons of AER gradients were made by using nonparametric tests because of the smaller sample size. The change in GFR was estimated by means of regression analysis of all available yearly GFR measurements and expressed as milliliters per minute per 1.73 square meters per year. Kruskal-Wallis test was used for multiple comparisons, and Wilcoxon's test was used for paired comparisons.

RESULTS

Table 1 lists baseline characteristics of the 33 patients included in the study and the 9 patients who were withdrawn. There were no major differences between the two groups. Table 2 lists the three treatment groups for the 33 patients included in the study. No differences in age, height,

Table 2. Baseline Characteristics of the Three Treatment Groups

Variable	Perindopril (n = 13)	Nifedipine (n = 10)	Placebo (n = 10)
Age (y)	35 ± 5	28 ± 4	28 ± 3
Diabetes duration (y)	21 ± 4	12 ± 2	15 ± 1
Sex distribution (M/F)	4/9	6/4	7/3
Smoking (%)	36	18	45
Weight (kg)	69 ± 5	71 ± 3	75 ± 4
HbA_{1c} (%)	8.5 ± 0.3	9.1 ± 0.7	9.2 ± 0.4
Total cholesterol (mmol/L)	4.9 ± 0.3	4.9 ± 0.4	5.0 ± 0.3
Triglycerides (mmol/L)	1.5 ± 0.2	1.2 ± 0.3	1.1 ± 0.2
Systolic BP (mm Hg)	133 ± 3	131 ± 3	132 ± 4
Diastolic BP (mm Hg)	80 ± 2	75 ± 2	77 ± 2
Albuminuria (μg/min)	66 ×/÷ 1.2	59 ×/÷ 1.2	66 ×/÷ 1.2
GFR (mL/min/1.73 m^2)	102 ± 6	105 ± 5	103 ± 8
Plasma creatinine (μmol/L)	87 ± 6	84 ± 6	83 ± 5

NOTE. Data shown as mean ± SEM, except for albuminuria (geometric mean ×/÷ tolerance factor). Treatment groups include patients completing the study.

weight, or duration of diabetes were observed among the three randomized groups.

Serial BP data are shown as mean arterial pressure (MAP) in Fig 1. There was a tendency for MAP to be lower in the perindopril-treated group, but this did not reach statistical significance for the study period as a whole. Comparisons at specific intervals showed a transient difference in MAP at 6 months between the two active drug therapies and placebo (perindopril, 90 ± 2 mm Hg [n = 13]; nifedipine, 90 ± 3 mm Hg [n = 10]; placebo, 98 ± 2 mm Hg [n = 10]; $P < 0.05$ versus placebo), whereas this same comparison showed a borderline statistically significantly difference at 12 ($P = 0.07$) and 24 months ($P = 0.06$). By 36 months of treatment, there was no statistically significant difference in MAP among the three treatment groups. During first 4 years of the study, hypertension developed in three patients in the placebo group. One of these patients was randomly allocated to therapy with nifedipine at 30 months, and the other two patients, to perindopril at 54 months of follow-up.

Twenty-four–hour ambulatory BP monitoring was assessed in all patients at least once during the active treatment phase over the study period. In this normotensive population, there was a slight decrease in ambulatory BP in the perindopril group that was more apparent at night (24-hour MAP: placebo, 96 ± 2 mm Hg; perindopril, 89 ± 1 mm Hg; nifedipine, 93 ± 3 mm Hg; awake MAP: placebo, 99 ± 2 mm Hg; perindopril, 94 ± 2 mm Hg; nifedipine, 96 ± 3 mm Hg; sleep MAP: placebo, 90 ± 3 mm Hg; perindopril, 81 ± 2 mm Hg; nifedipine, 87 ± 4 mm Hg; $P < 0.05$, perindopril versus placebo).

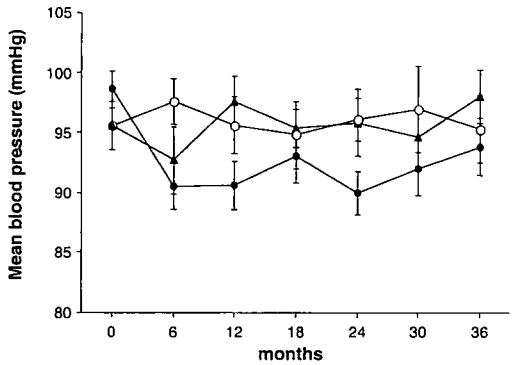

Fig 1. Data shown as mean ± SEM for MAP over the 36-month study period. (●), Perindopril treatment; (○), placebo; (▲), nifedipine treatment.

No changes in glycosylated hemoglobin level were observed with any treatment (mean HbA_{1c} level on treatment: perindopril, 8.9% ± 0.2%; nifedipine, 9.1% ± 0.4%; placebo, 9.2% ± 0.5%; normal range, 4.3% to 6.1%). The plasma renin concentration was increased in the perindopril group at 36 months ($P < 0.01$) compared with both the nifedipine and placebo groups (perindopril, 5.6 ± 1.0 ng/mL/h; nifedipine, 1.7 ± 0.3 ng/mL/h; placebo, 1.9 ± 0.4 ng/mL/h; normal range, 1 to 3 ng/mL/h; $P < 0.05$ for perindopril versus other groups).

There were no significant differences among the three treatment groups in regard to serum cholesterol and triglyceride levels at any time over the study period (mean data over 36 months: serum cholesterol, perindopril, 5.2 ± 0.3 mmol/L (n = 11); nifedipine, 4.9 ± 0.2 mmol/L (n = 9); placebo, 5.2 ± 0.3 mmol/L (n = 11); serum triglycerides, perindopril, 1.4 ± 0.2 mmol/L; nifedipine, 1.5 ± 0.5 mmol/L; placebo, 1.5 ± 0.2 mmol/L).

Major findings relate to the effects of the three treatments on AER. Baseline AER was in the low microalbuminuric range (geometric mean: perindopril, 66 μg/min; nifedipine, 59 μg/min; placebo, 66 μg/min), with no significant differences among the three groups (Table 2). In the placebo group, AER continued to progressively increase over the 3-year study period. A similar increase in AER was observed with nifedipine treatment. Conversely, perindopril therapy was associated with an initial decline in AER (Fig 2). Median AERs at 3 years of treatment in each group were 23 μg/min for perindopril, 122 μg/min for nifedipine, and 112 μg/min for placebo (Fig 2; $P < 0.01$). To further assess the effect of the treatments on AER, the yearly rate of change in AER was determined for each patient (Fig 3). In the majority of the perindopril-treated patients, there was a decline in AER, whereas in the other groups, most patients had an increase in AER ($P < 0.03$; Fig 3). Further analysis was performed to determine the number of patients who progressed or regressed within each study group (Table 3). Of the 13 perindopril-treated patients, only 1 patient developed macroalbuminuria, whereas in 7 patients, AER regressed to less than

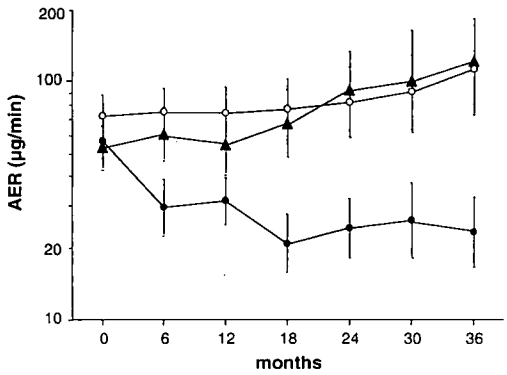

Fig 2. Data shown as geometric mean ×/÷ tolerance factor on a logarithmic scale for AER over the 36-month treatment period. $P < 0.01$ at 36 months for perindopril versus placebo and nifedipine. $P < 0.02$ for perindopril versus nifedipine at 24 months. (●), Perindopril treatment; (○), placebo; (▲), nifedipine treatment.

20 µg/min compared with no patients regressing to normal in the nifedipine-treated or placebo groups (perindopril versus placebo, $P < 0.01$; perindopril versus nifedipine, $P < 0.001$, Fisher's exact test; Table 3).

To separately examine the short-term and long-term effects of each treatment on AER, the percentage of change in AER per year in individual patients was calculated for the first 12 months of the study and from 12 months to the final data collection point for the 25 patients with data beyond 36 months. In the first 12 months, the

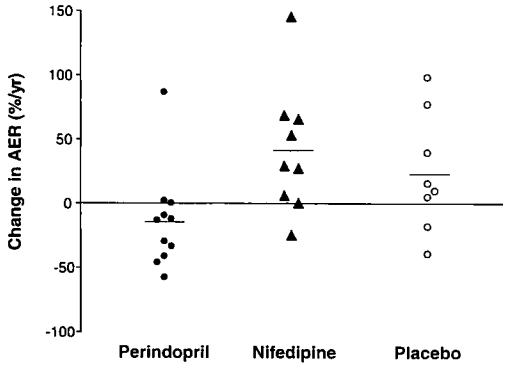

Fig 3. Data shown for individual rates of change in albumin excretion per year. Horizontal lines represent the mean change for each treatment group. Analysis of variance, perindopril versus nifedipine, $P < 0.03$; perindopril versus placebo, $P = 0.08$; and nifedipine versus placebo, P = NS.

Table 3. Status of Albuminuria in the Three Treatment Groups up to 36 Months of Follow-Up

Variable	Perindopril (n = 13)	Nifedipine (n = 10)	Placebo (n = 10)
Regression to normoalbuminuria	7 (54)	0* (0)	0* (0)
Microalbuminuria	5 (38)	6 (60)	7 (70)
Progression to macroalbuminuria	1 (8)	4 (40)	3 (30)

NOTE. Values expressed as number (percent).
* $P < 0.01$ versus perindopril, Fisher's exact test.

median reduction in AER in the perindopril group was 45%. However, AER increased by 17.6% in the nifedipine group and 27.6% in the placebo group ($P < 0.03$, Kruskal-Wallis). After the first 12 months, there was a significant attenuation in the median rate of decline in AER in the perindopril group from 45% in the first year to 12.5% per year beyond 12 months ($P = 0.02$). Conversely, there was no significant difference in the rates of increase in AER after the first 12 months in either the nifedipine group (+17.6% initial versus +22.8% per year after 12 months; P = not significant [NS]) or placebo group (+27.6% initial versus +9.6% per year after 12 months; P = NS). In the perindopril group, there was a significant attenuation in the rate of decline of AER after 12 months. Conversely, there was no significant change in AER gradients in the nifedipine group ($P = 0.21$) or placebo group ($P = 0.99$). Individual AER data for the three treatment groups are shown in Fig 4 according to duration of follow-up. A pooled analysis of results at month 24 in patients from all three treatment groups showed a significant correlation between the gradient in AER in individual patients and the change in systolic BP ($r = 0.48$; $P < 0.01$) and diastolic BP ($r = 0.34$; $P = 0.05$).

Three-month BP data after cessation of antihypertensive therapy were available for 14 patients (8 patients, perindopril; 6 patients, nifedipine). Patients in either the perindopril or nifedipine subgroups did not show a significant change in systolic BP, diastolic BP, or MAP after 3 months off treatment. Three-month AER data off antihypertensive therapy are shown in Fig 5. In perindopril patients, AER increased from 20 to 47 µg/min at 3 months off treatment ($P < 0.001$). In nifedipine patients, AER levels remained el-

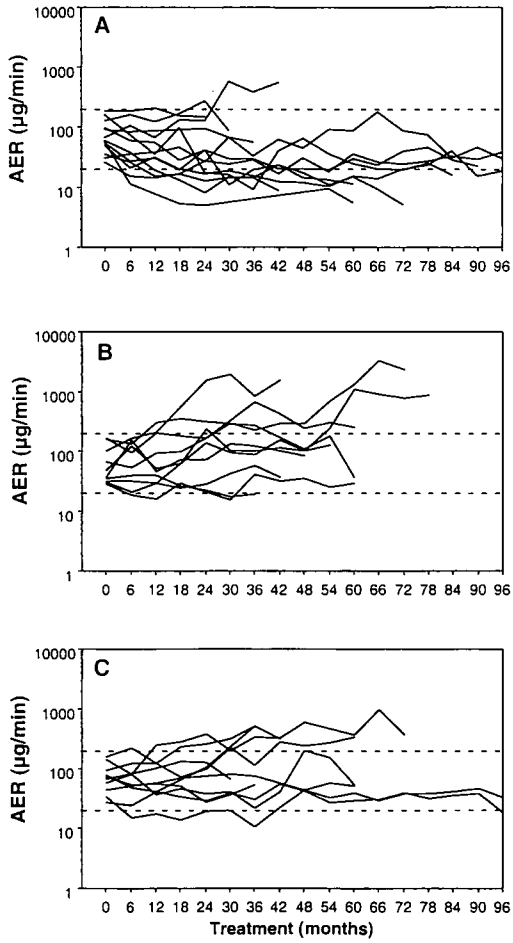

Fig 4. Individual AER data are shown for the (A) perindopril, (B) nifedipine, and (C) placebo groups according to duration of follow-up.

evated but did not increase further after cessation of therapy (on therapy, 130 μg/min versus off therapy, 132 μg/min; P = NS). In the 8 perindopril patients, there was no relationship between study duration and the on-off treatment increment in AER ($r = 0.18$; $P = 0.69$).

Baseline GFR was similar in the three treatment groups (perindopril, 102 ± 6 mL/min/1.73 m^2; nifedipine, 105 ± 5 mL/min/1.73 m^2; placebo, 103 ± 8 mL/min/1.73 m^2; $P = 0.95$). There was no significant change in GFR in any group over 3 years. In the 25 patients with data beyond 3 years, the rate of decline in GFR (in milliliters per minute per 1.73 square meters per year) was calculated for each individual (Table 4). In the nifedipine group, GFR declined more rapidly than in the other two groups, and neither the perindopril group nor the placebo group showed a significant change in GFR from baseline. One subject each in the perindopril and nifedipine groups showed a decline in GFR to less than 60 mL/min/1.73 m^2 after 36 months.

DISCUSSION

The principal finding in this study is that the increase in urinary albumin excretion in normotensive microalbuminuric patients with type 1 diabetes was attenuated during 3 to 8 years of perindopril therapy. Conversely, nifedipine treatment in these patients was associated with a progressive increase in AER similar to that observed in the placebo group. In subjects studied longer than 3 years, AER continued to increase in the nifedipine and placebo groups, and GFR declined significantly in the nifedipine group. However, in the perindopril group, median AER was reduced by 45% in the first year, and this rate of decline was attenuated to 12.5% per year thereafter. The difference in albuminuria response between perindopril and nifedipine patients was large, whereas both drugs had only modest effects on BP in these normotensive patients. Nevertheless, one cannot exclude the possibility that small changes in attained BP

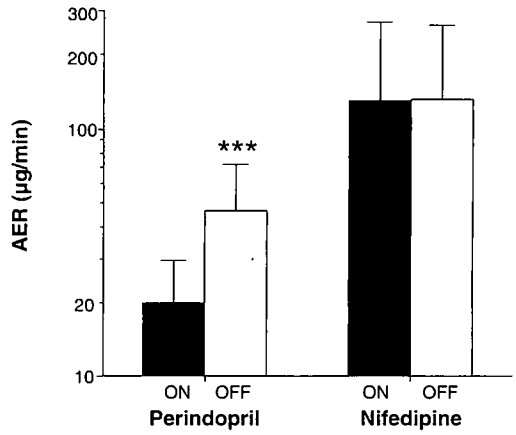

Fig 5. Data shown as mean ± SEM on a logarithmic scale for AER measurements performed on the last sample on treatment and then off treatment. Three-month AER data off antihypertensive therapy were available for 14 patients (8 patients, perindopril; 6 patients, nifedipine). ***P < 0.001, ON versus OFF in perindopril group.

Table 4. GFR in 25 Subjects With Data Beyond 36 Months

	Perindopril (n = 9)	Nifedipine (n = 9)	Placebo (n = 7)	*P*
Duration of follow-up (mon)	67 ± 8	51 ± 4	65 ± 8	0.17
Initial GFR (mL/min/1.73 m^2)	95 ± 7	110 ± 6	90 ± 7	0.11
Final GFR (mL/min/1.73 m^2)	90 ± 9	80 ± 5	82 ± 5	0.60
Change in GFR/y (mL/min/1.73 m^2)	−1.0 ± 1.2	−7.8 ± 1.8	−1.3 ± 1.1	0.004

GFR is shown as mL/min/1.73 m^2.

could have influenced the results. Although MAP was not significantly different among the three treatment groups for the study period as a whole, MAP in the perindopril group was at least 2 mm Hg lower than in the nifedipine or placebo groups at some time points (Fig 1).

Several studies have shown that ACE inhibitors reduce AER or attenuate increases in AER in normotensive microalbuminuric patients over 1,[2] 2,[4,17,18] or 4 years.[1] However, comparisons between ACE inhibitors and placebo have been unable to resolve the separate contributions of a reduction in systemic BP and an intrarenal effect on the reduction in AER. A three-arm study including an ACE inhibitor, another antihypertensive agent used in equihypotensive doses, and a placebo is required to increase our understanding of the effects of ACE inhibition in evolving diabetic nephropathy.

Only one study used a three-arm design and directly compared a dihydropyridine CCB with an ACE inhibitor in normotensive patients with type 1 diabetes with microalbuminuria.[19] In that study, the effects of lisinopril, nifedipine, and placebo therapy were assessed over 3 years in 92 patients. Both lisinopril and nifedipine attenuated increases in albuminuria. The only parameter supporting a greater attenuation of AER by lisinopril than nifedipine was a smaller proportion of lisinopril patients with a greater than 50% yearly increase in AER. However, because the lisinopril group attained a lower MAP than the nifedipine or placebo groups, it is not clear whether lisinopril exerted an antialbuminuric effect independent of a reduction in systemic BP.

The first report by the MDNSG was unable to detect significant differences in AERs when perindopril was compared with nifedipine therapy in hypertensive and normotensive microalbuminuric patients with type 1 and type 2 diabetes over 12 months.[7] In the present study, AER gradients were similar in the perindopril and nifedipine groups over the first 12 months, with a difference emerging only after 24 to 36 months. However, the lack of a concurrent placebo group in the initial study and different criteria for classification of hypertension and diabetes type complicates comparisons of the present data with those from the first study.

The perindopril- and nifedipine-treated groups had lower levels of diastolic BP than the placebo group at 6 months. This was maintained in the perindopril group, whereas an initial decrease was followed by a gradual increase in diastolic BP after the titration phase in the nifedipine group. When a pooled analysis was performed on patients from all three treatment groups, there was a significant correlation between AER gradient in individual patients and change in systolic and diastolic BPs. The difference in attained BP between perindopril and nifedipine treatment was not significant and was less than 4 mm Hg. However, this does not exclude a role for systemic BP as a potential explanation for the difference in effect of the two drugs on AER in this study.

A correlation between changes in BP and changes in AER was observed in the present study and was also shown in the initial MDNSG study.[7] These findings and those of Crepaldi et al[19] imply that changes in systemic BP may modulate AER even when BP levels are in the normotensive range and that any intrarenal effects of specific antihypertensive agents are superimposed on this.

No significant changes in GFR were observed among the three groups during the first 3 years. In the 25 patients followed up for longer than 3 years, there was no significant change in GFR in the perindopril and placebo groups. This is consistent with the concept that in patients with type 1 diabetes, GFR does not show a persistent decline in patients with stable microalbumin-

uria.[20] Conversely, GFR declined by approximately 8 mL/min/1.73 m^2/y in nifedipine-treated patients followed up longer than 3 years, although final attained GFR did not differ among the three treatment groups. A similar effect of nifedipine on renal function has been reported previously in patients with more advanced renal disease.[21,22] A recent study compared enalapril with nifedipine therapy in hypertensive patients with type 2 diabetes with varying degrees of albuminuria over a mean duration of 5 years.[22] Renal function was assessed by the regression coefficients of the reciprocal of plasma creatinine measured yearly in all patients. In 34 microalbuminuric patients, there was no significant change in renal function overall and no difference between the enalapril and nifedipine groups. Conversely, in 25 macroalbuminuric patients, renal function declined in the nifedipine group but was stabilized by enalapril.[22] In a meta-analysis of studies of antihypertensive therapy in patients with overt diabetic nephropathy, a decline in GFR of 42% per year (95% confidence interval, 8 to 76) was shown in groups treated with nifedipine compared with no change in groups treated with other agents, including ACE inhibitors.[21]

The lack of effect of the dihydropyridine CCB on AER despite similar effects on systemic BP is consistent with previous experimental and clinical studies. In the rodent, several studies have indicated in both hypertensive and normotensive models that despite equivalent BP reduction, dihydropyridine CCBs do not exert the renoprotective actions observed with ACE inhibitors.[5,23] A possible deleterious effect of CCBs on the kidney has been recently confirmed in the experimental model of renal ablation,[24] a model with many hemodynamic and structural similarities to diabetic nephropathy.[25] The differences in renal protection between the two treatment regimens may be related to differences in the mode of action of these agents within the kidney. Experimental data suggest differences in effects on such intrarenal hemodynamics as intraglomerular pressure,[26] and it is possible that these agents have disparate effects on renal cytokine expression.[27]

Most clinical studies evaluating CCBs in diabetic nephropathy have focused on subjects with type 2 diabetes.[28,29] In such studies, dihydropyridine CCBs have generally failed to be as effective as ACE inhibitors.[29-31] However, it is important to note that the results obtained in the present study may not apply to such newer formulations as the controlled-release form of nifedipine, which has better 24-hour availability.

One previous study of 44 normotensive microalbuminuric subjects with type 1 diabetes has shown that decreases in albuminuria associated with ACE inhibitor therapy were subsequently coupled with the prevention of a decline in GFR after 8 years of follow-up.[20] Even in that study, no difference could be detected in GFR between patients administered an ACE inhibitor and patients not on antihypertensive therapy when a group comparison was performed. In that study, fewer patients treated with an ACE inhibitor developed macroalbuminuria over 8 years, and only those patients developing macroalbuminuria showed a decline in GFR.[20] A relationship between treatment-induced reduction in albuminuria and subsequent decline in GFR has been more readily detected in patients with[32] and without diabetes[33] with advanced nephropathy.

It remains to be determined whether the functional effects of ACE inhibition observed in this study are associated with protective effects on renal structure. Experimental studies have indicated that ACE inhibition retards glomerular basement membrane thickening[34] and tubulointerstitial injury[27] in normotensive diabetic rats. In two studies of humans, perindopril was associated with the preservation of renal and, in particular, tubulointerstitial morphological states in patients with diabetes with renal disease.[35,36] In a small study of normotensive microalbuminuric patients with type 1 diabetes, an ACE inhibitor and a β-blocker were shown to arrest development of glomerular ultrastructural injury to a similar extent over 3 to 4 years,[37] indicating that hypotensive agents other than ACE inhibitors may exert beneficial effects on renal structure.

In the present study, a significant number of patients remained normotensive despite cessation of antihypertensive therapy. This made it possible to determine whether there was a rapid increase in AER after ceasing active therapy, as observed previously in the first MDNSG study[7] and by Passa et al.[38] Cessation of ACE inhibition

was associated with an approximate doubling in AER over 3 months in patients previously treated with perindopril, but a similar increase in AER was not observed with cessation of nifedipine (Fig 4). The modest yet rapid increase in AER after ceasing perindopril therapy may represent a failure of ACE inhibitors to completely inhibit the underlying structural damage related to the nephropathic process. Nevertheless, the level of albuminuria in this group was still lower than that seen in the placebo or nifedipine-treated groups after cessation of active treatment.

This study has documented three aspects of long-term ACE inhibition on albuminuria in normotensive microalbuminuric patients with type 1 diabetes: (1) a marked reduction in AER in the first 12 months of therapy, (2) attenuation of the rate of decline of AER during long-term follow-up, and (3) increase in AER after short-term cessation of perindopril therapy. This pattern of AER response was not observed with nifedipine. Furthermore, nifedipine therapy was associated with a progressive decline in GFR.

The present study indicates that in microalbuminuric patients with type 1 diabetes with normotension, ACE inhibitors are more effective than dihydropyridine CCBs in retarding the long-term progression of albuminuria. These findings extend the rationale for the use of ACE inhibitors as first-line therapy in normotensive microalbuminuric patients with type 1 diabetes.

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APPENDIX

Melbourne Diabetic Nephropathy Study Group II

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